



CS 431/631 Introduction to Big Data

Introduction to Different Learning Types

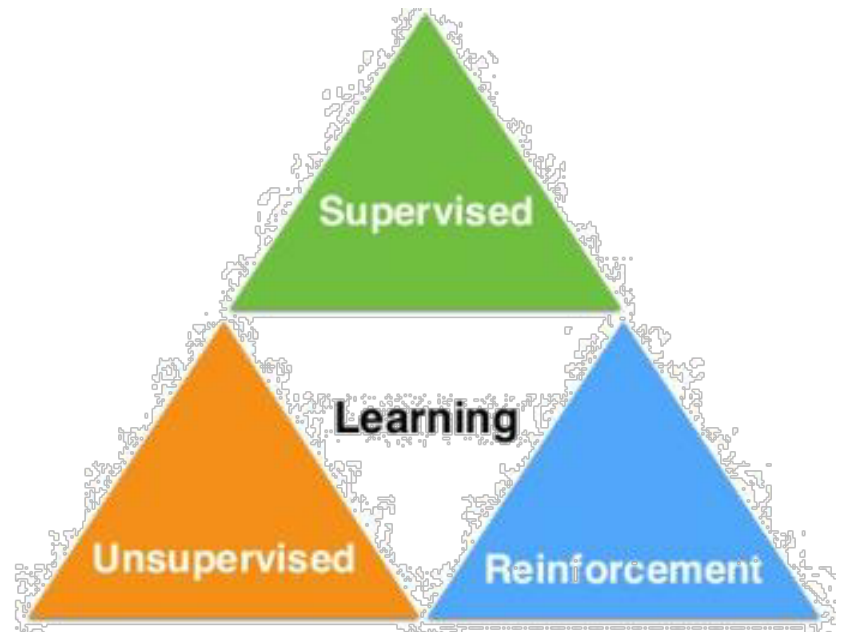
Some slides are borrowed from Yiran Chen, Hai Li, et. al.

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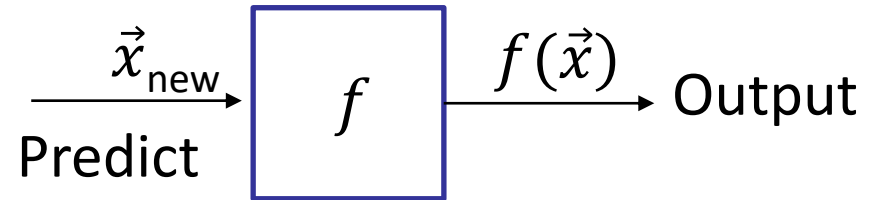
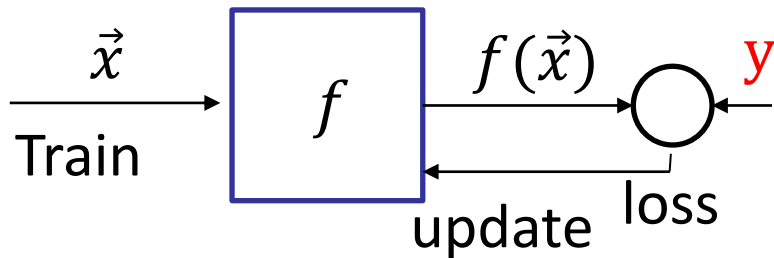
The overview of learning

- People consider learning as finding a function.
- Supervised learning:
 - $f(\text{input}) \rightarrow \text{target}$ (continuous/discrete)
- Unsupervised learning:
 - $f(\text{input}) \rightarrow \text{likelihood}$
- Reinforcement learning:
 - $f(\text{state, action}) \rightarrow \text{value}$



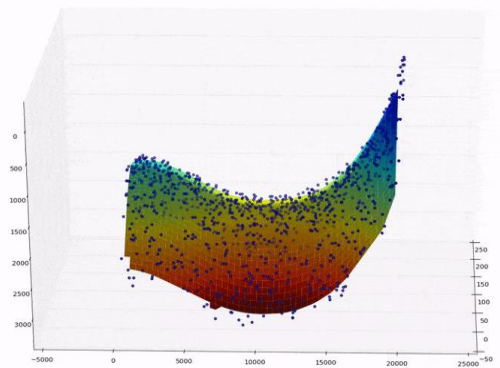
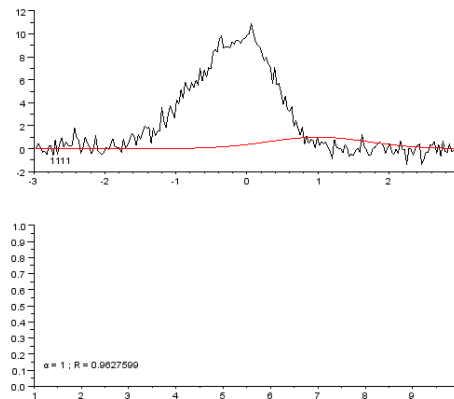
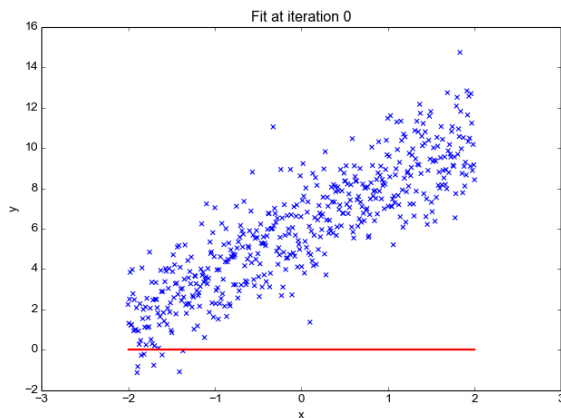
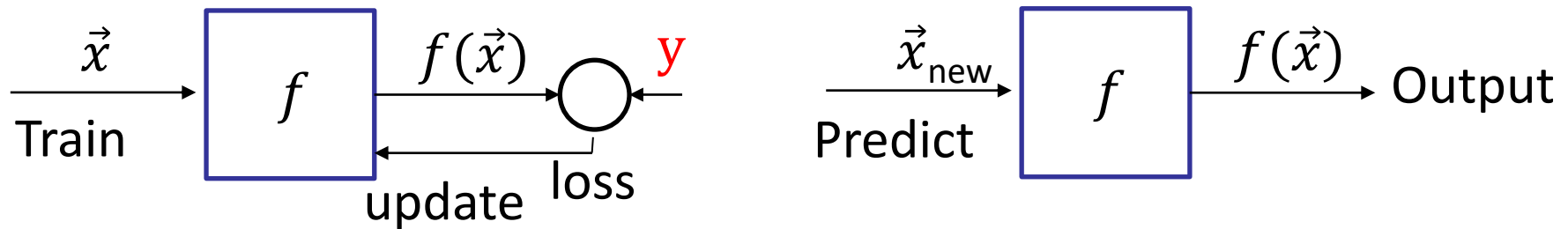
Supervised learning

- Function: $f(\text{input}) \rightarrow \text{target}$
 - Continuous value: Regression
 - Discrete value: Classification



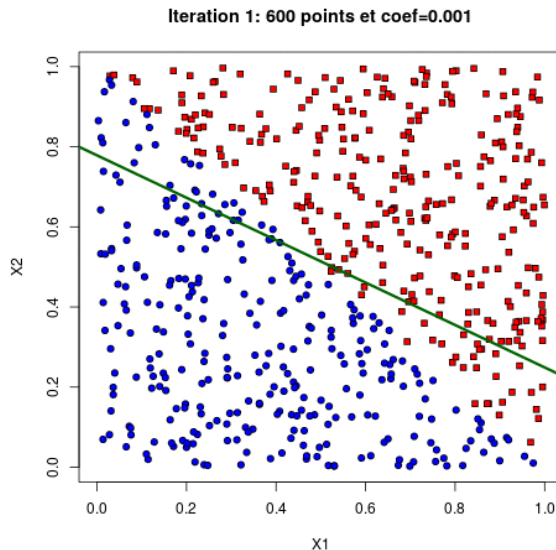
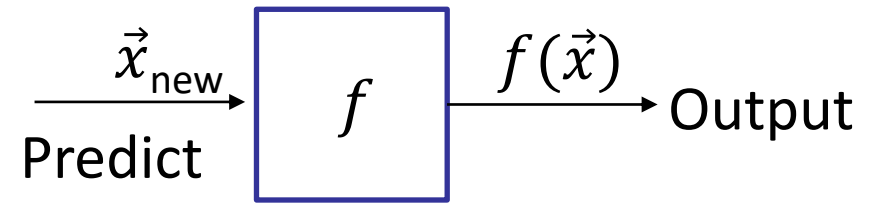
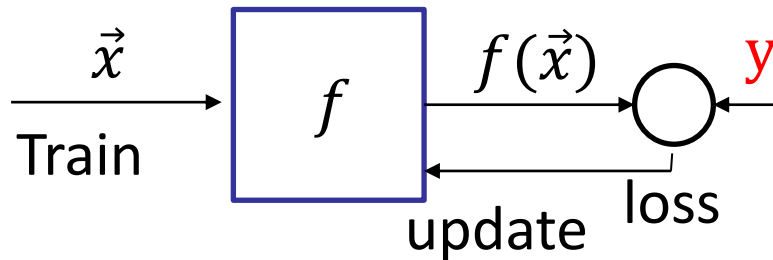
Regression

- Use data to identify relationships among variables and then apply these relationships to predict.
- Continuous output.

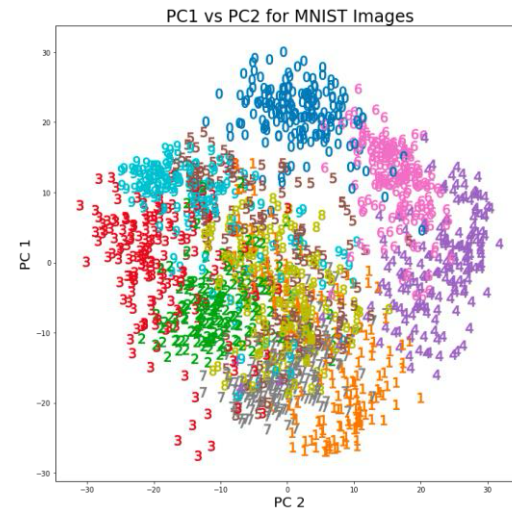


Classification

- Similar with regression.
- The only different: Label y & output are discrete.



Linear



Non-Linear

Unsupervised learning

- Function: $f(\text{input}) \rightarrow \text{likelihood}$
 - Clustering
 - Generative model

Clustering

- Cluster analysis or clustering is the task of
 - grouping a set of objects
 - in such a way that objects in the **same group** (called a cluster) are more **similar** (in some sense) to each other than to those in other groups (clusters).

Clustering: K-Means

- Input: Data set $x_1, \dots, x_n$, number of clusters K
- Output: Cluster centers $c_1, \dots, c_K$
- Goal: Minimize $\text{cost}(c_1, \dots, c_K) = \sum_i \min_k (\text{dist}(x_i, c_k))$

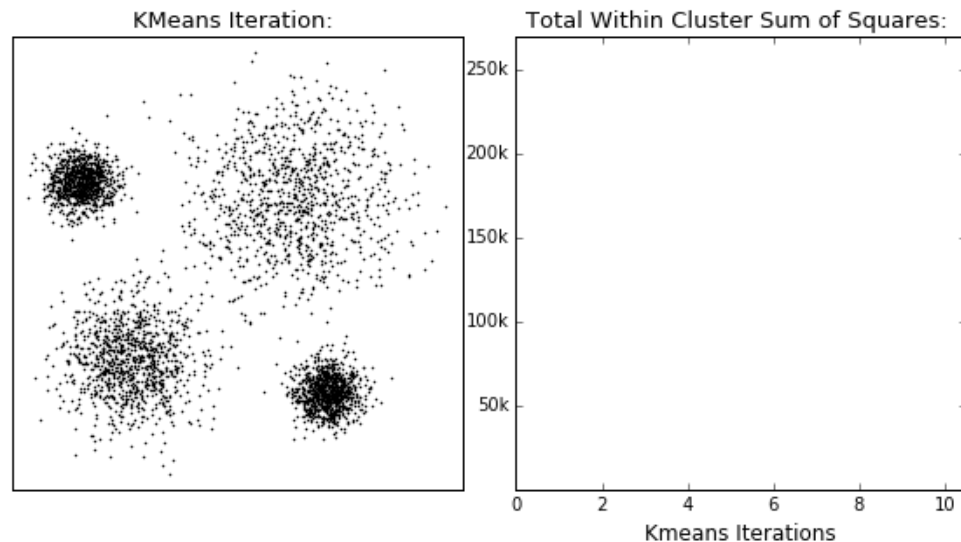
Pseudo code:

1. Randomly initial K centers.

Until converged:

2. Assign each data to its closest center.

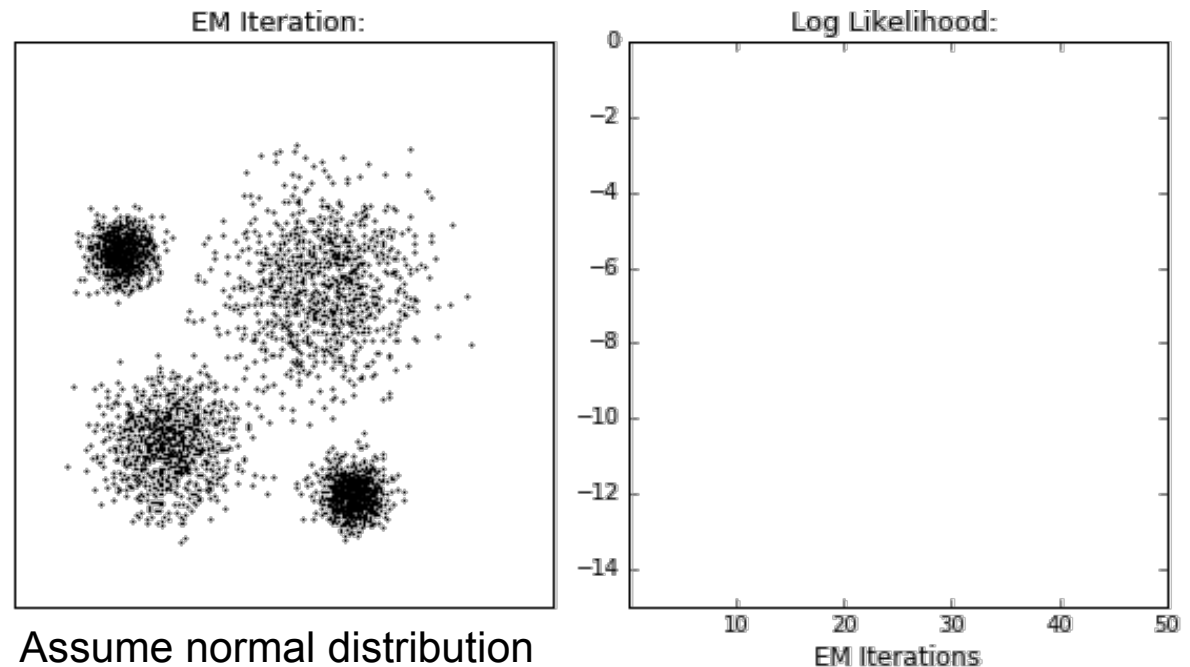
3. Move the original center to the average of all data belonging to it.



- K-Medoids (use data points), K-Medians (use median)

Clustering: Gaussian mixture model (GMM)

- Traditional method: Gaussian mixture model
 - Initialize distribution (map point to distribution)
 - Compute expectation
 - Expectation Maximization (EM)
 - Iterate to step 2



Generative Model

- Given training data, generate new samples from the same distribution.
- Traditional Method: Naïve Bayes
- Naïve Bayes assumption: features are **independent**.

$$p(X) = p(x_1, \dots, x_k) = \prod_{k=1}^K p(x_k)$$

- The problem is reduced to estimating the parameters' $p(x_k)$ for each feature separately and multiplying these to find the probability for any possible combination. It **cannot** generate reasonable new images with high-dimensional information.

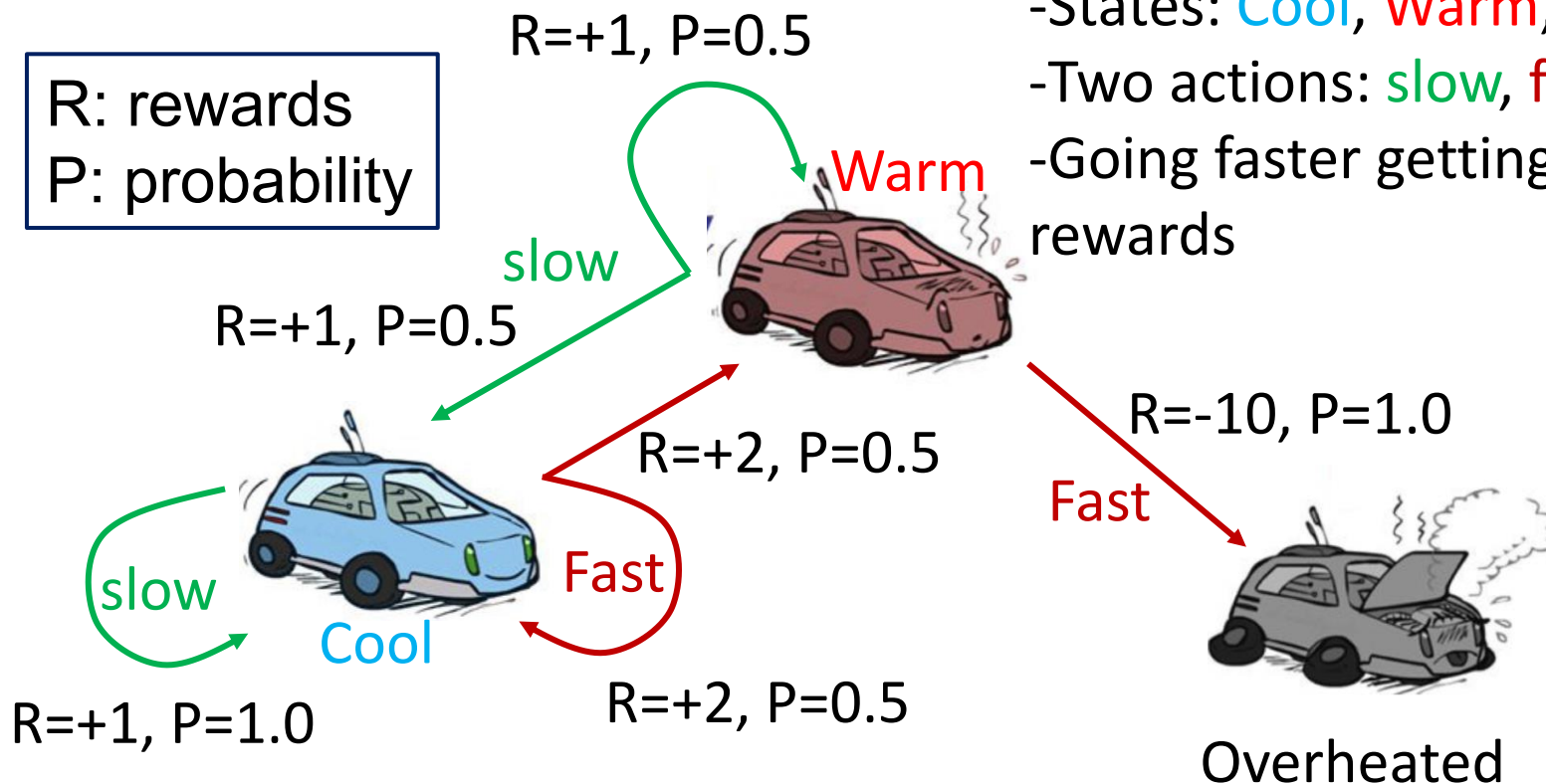


Fail on pixel cartoon image generation using naïve Bayes

Reinforcement learning

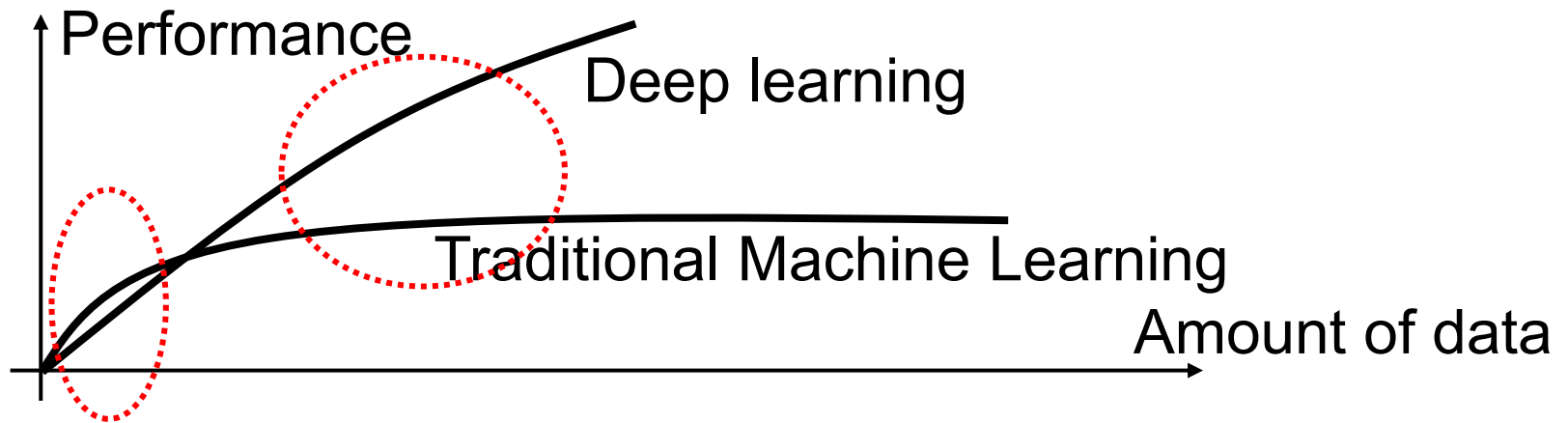
- Function: $f(\text{state}, \text{action}) \rightarrow \text{reward value}$
- Basic reinforcement is modeled as a Markov decision process.

-Example: car racing
-States: **Cool**, **Warm**, Overheated
-Two actions: **slow**, **fast**
-Going faster getting double rewards



Why do we need Deep Learning

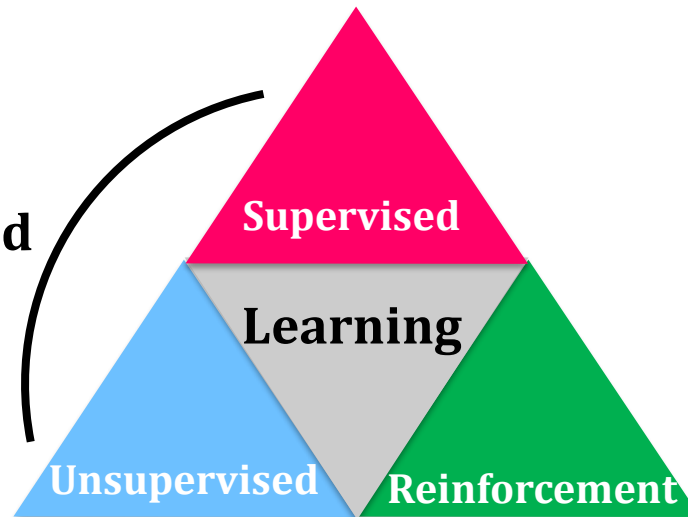
- In the real world.
 - Input data is high-dimensional
 - Mapping function is complicated
 - Traditional methods usually fail
- Deep Neural Network is capable of fitting functions with high complexity.



Learning Paradigms

Labeled data
Direct feedback
Predict outcome/future
Apps: Classification

Semi-supervised
Weakly-supervised



No labels
No feedback
Find hidden representations
Apps: Reconstruction

Decision process
Reward system
Learn series of actions
Apps: Decision-making

Supervised Learning

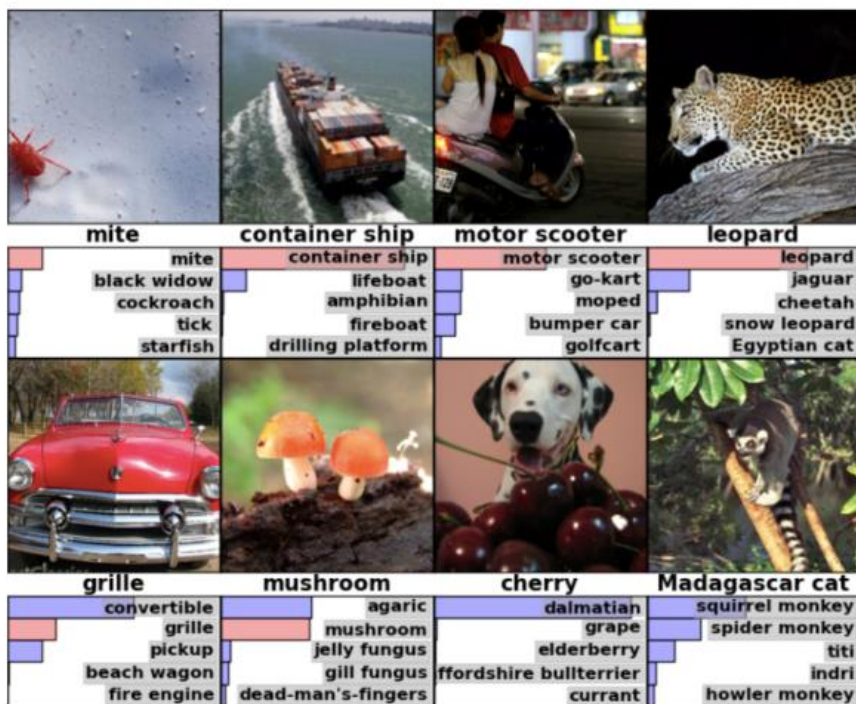
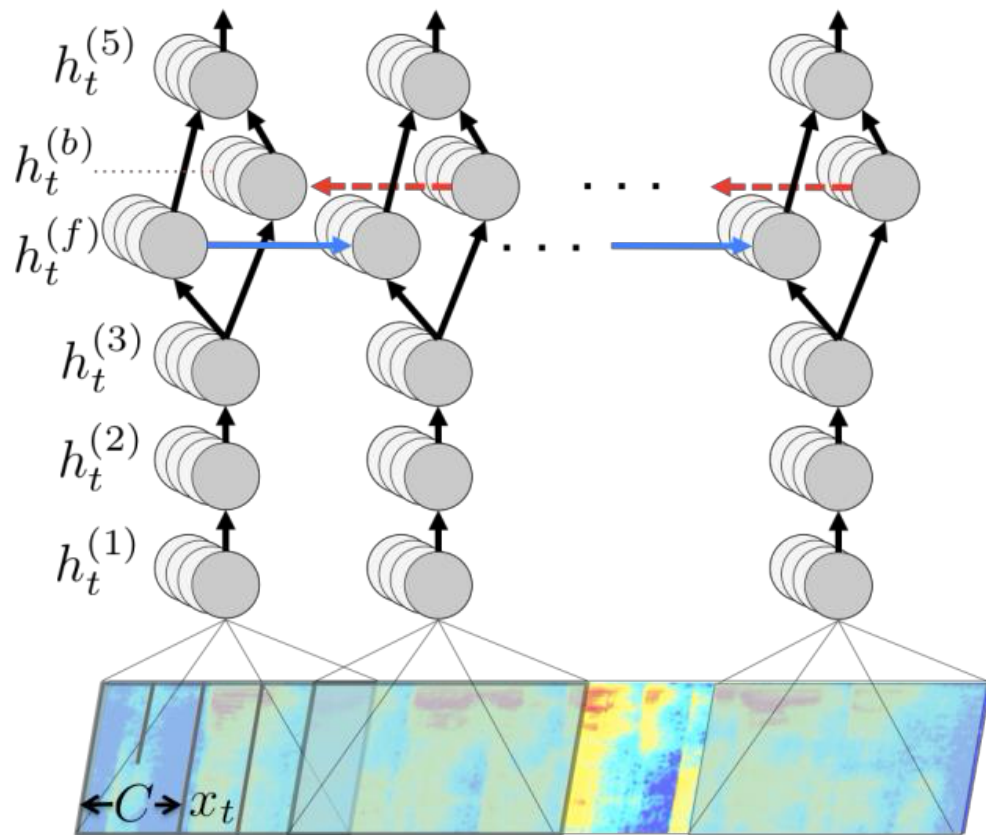


Image Classification



Speech Recognition

DL for supervised learning

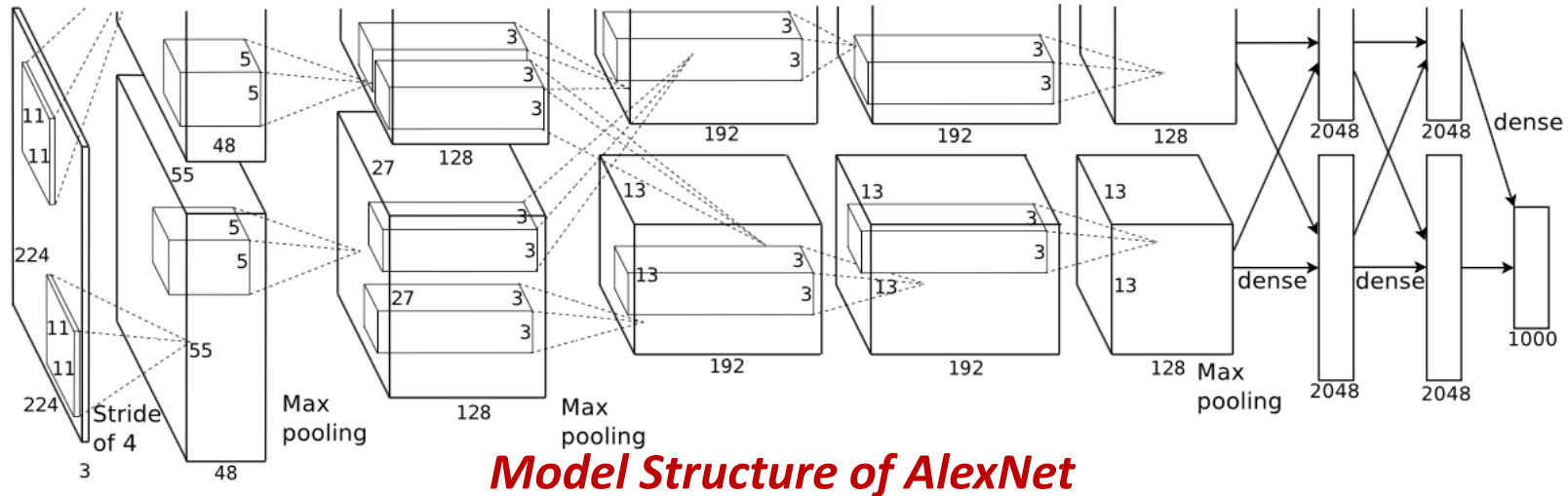
- Convolutional neural network
 - The network employs a mathematical operation called convolution.
 - Convolution is a specialized kind of linear operation.
- Structured learning
 - Labels are structured high dimensional data.

Compare to traditional ML methods

- Convolutional neural network
 - Traditional ML has the best score 26.2%.
 - ResNet-152 reached 3.57% error rate (ILSVRC'15).
 - Human's error rate is 5%.
- Structured learning
 - Most of these problems cannot be solved using traditional ML methods.

Convolutional neural network

Image Classification



Model Structure of AlexNet

Model	Top-1 Error	Top-5 Error
Sparse coding	47.10%	28.20%
SIFT + FVs	45.70%	25.70%
AlexNet	37.50%	17.00%

Performance Evaluation



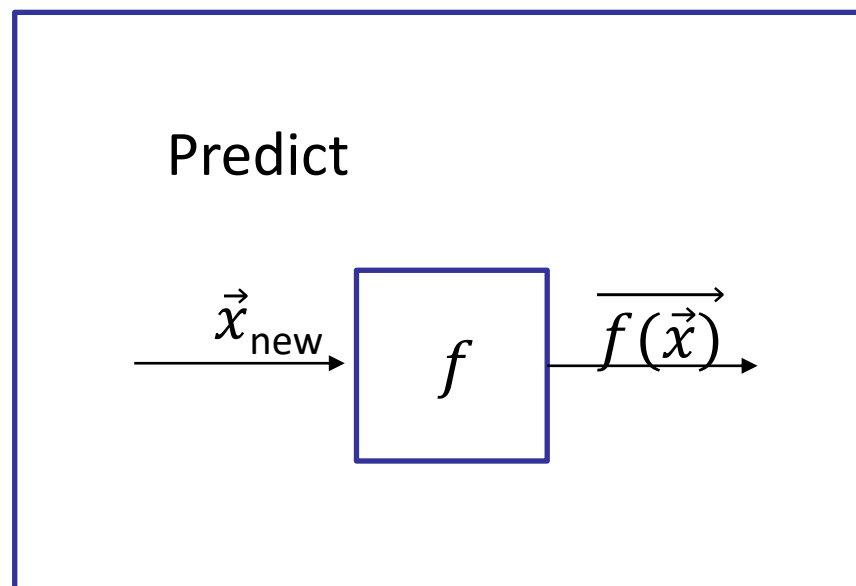
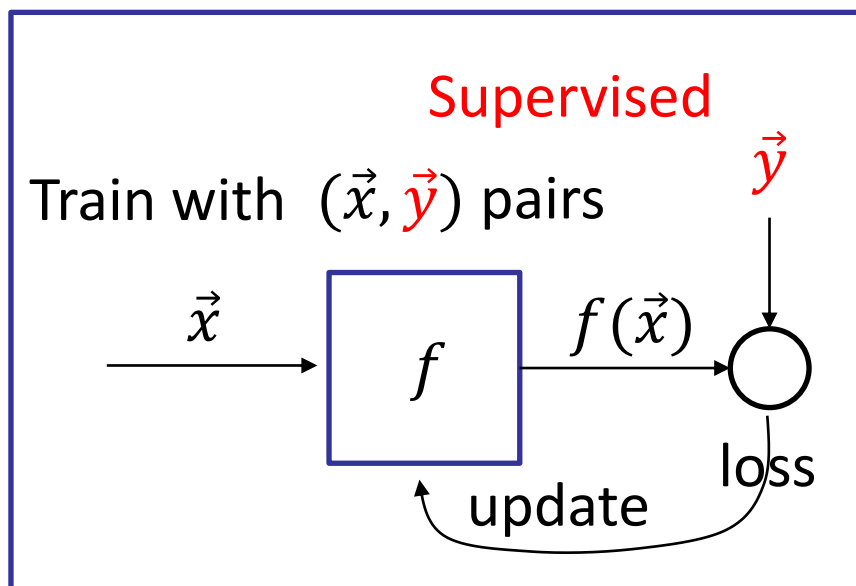
Sample Testing Result from ImageNet

SIFT: Scale Invariant Feature Transform, FVs: Fisher Vectors

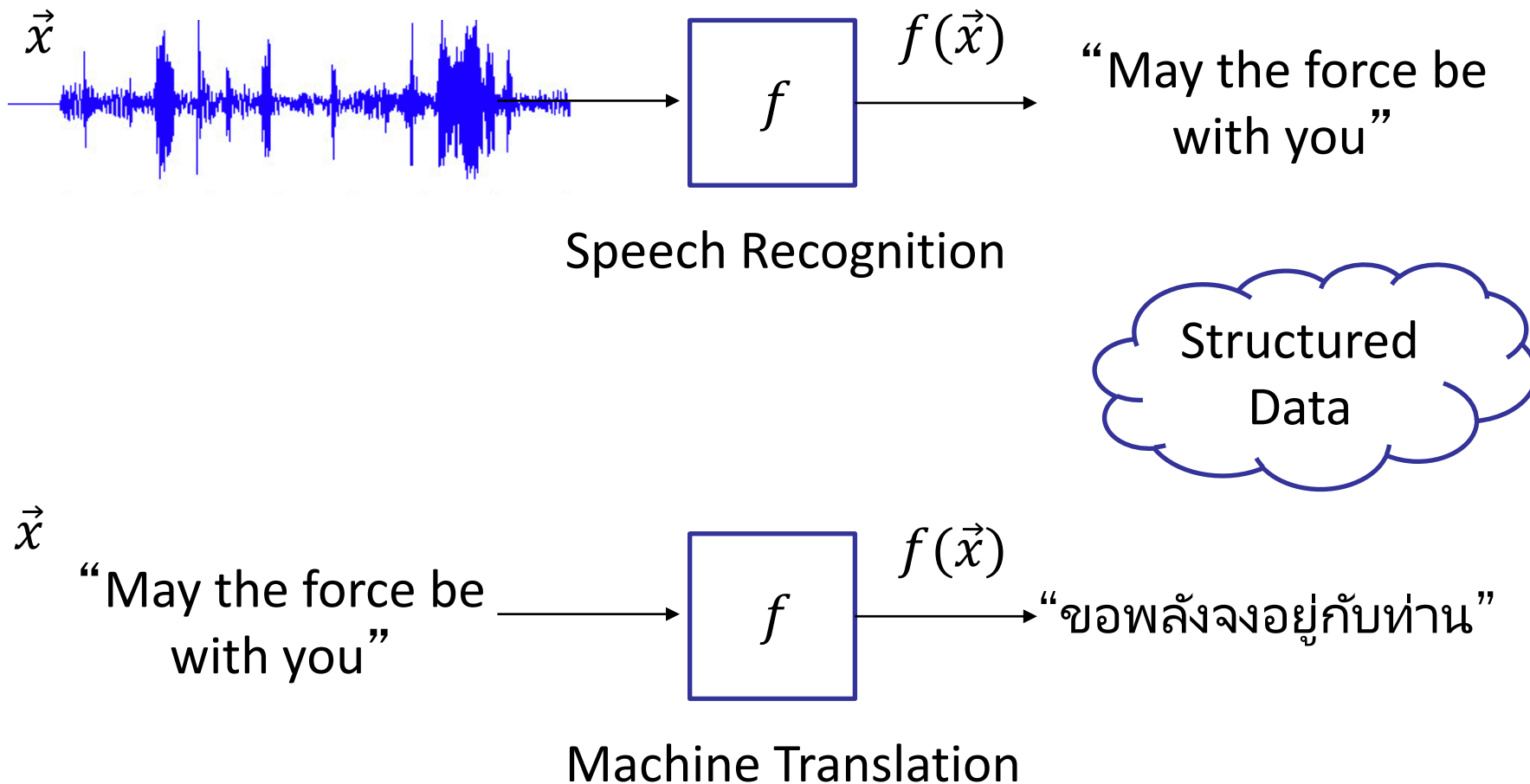
A. Krizhevsky, I. Sutskever and G. E. Hinton. ImageNet Classification with Deep Convolutional Neural Networks. NIPS, 2012.

Structured learning

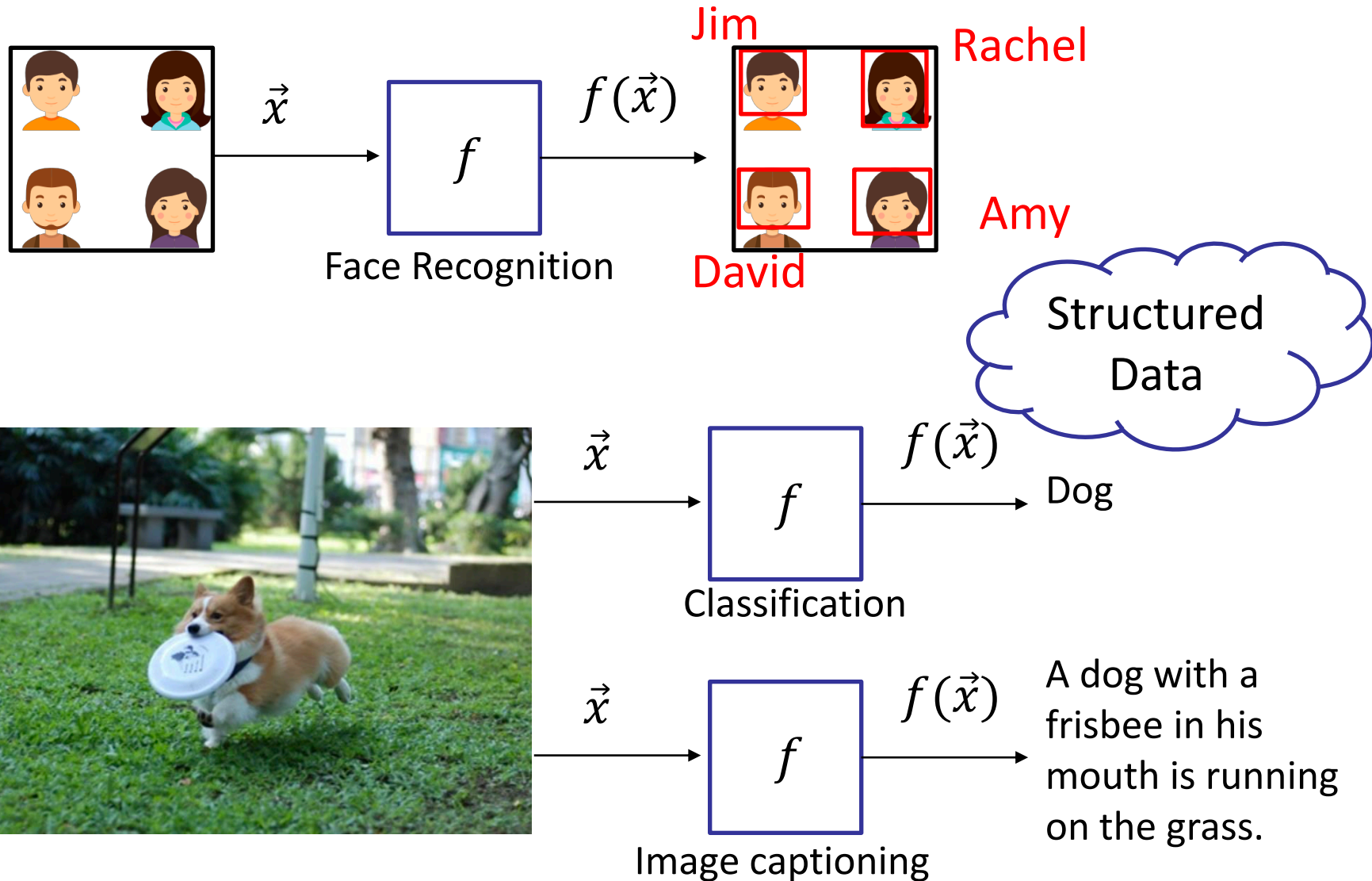
- Similar with regression and classification.
- The only difference is: Label y & output are structured data.
 - For example: image, sound, text...



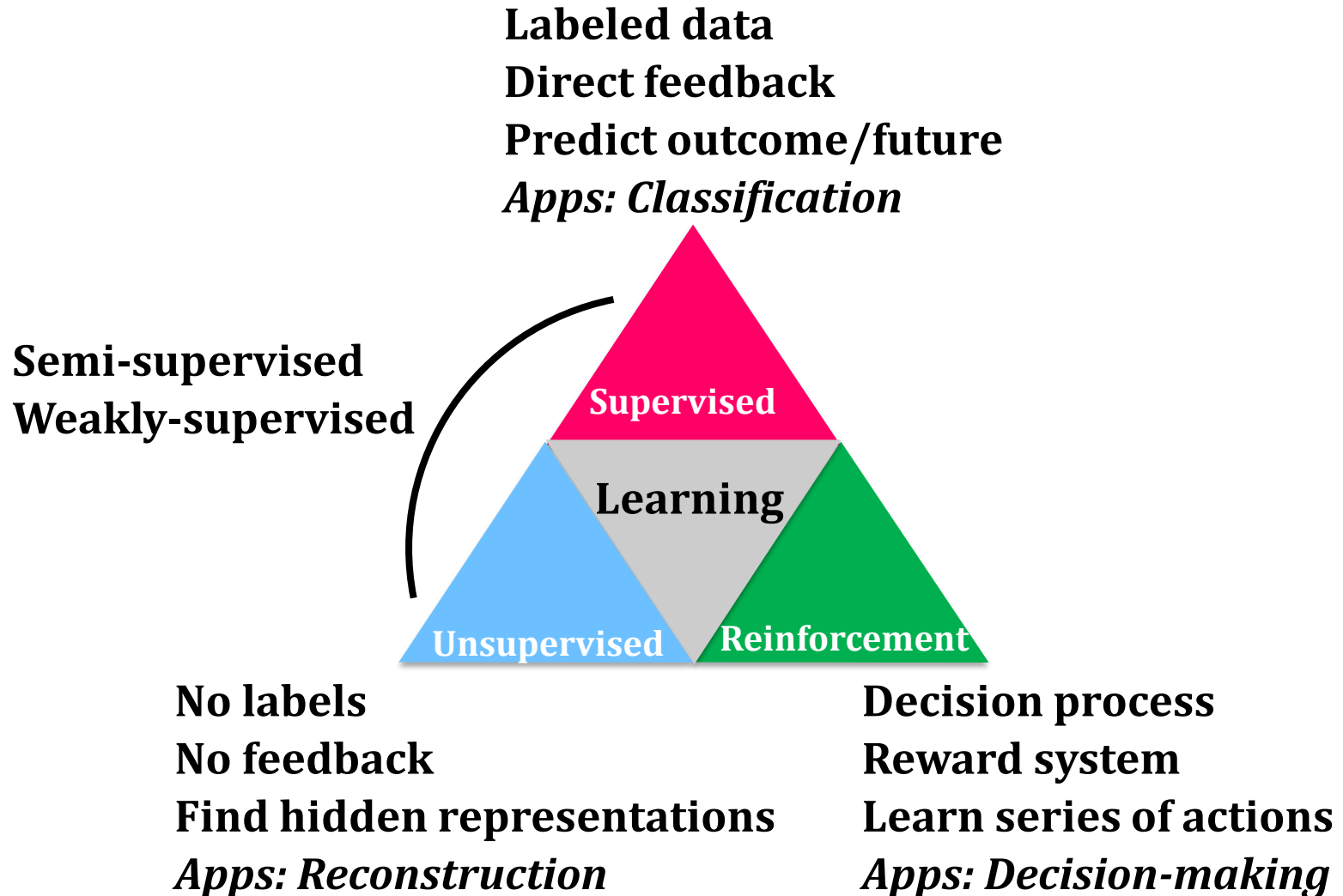
Structured learning



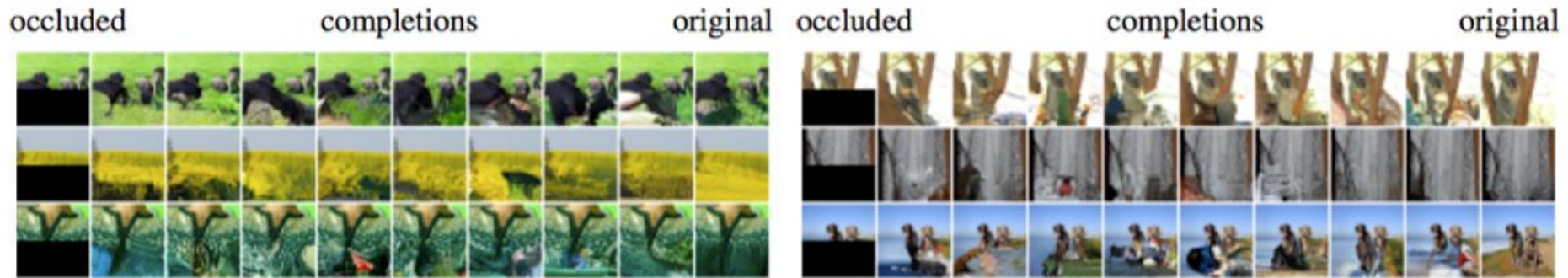
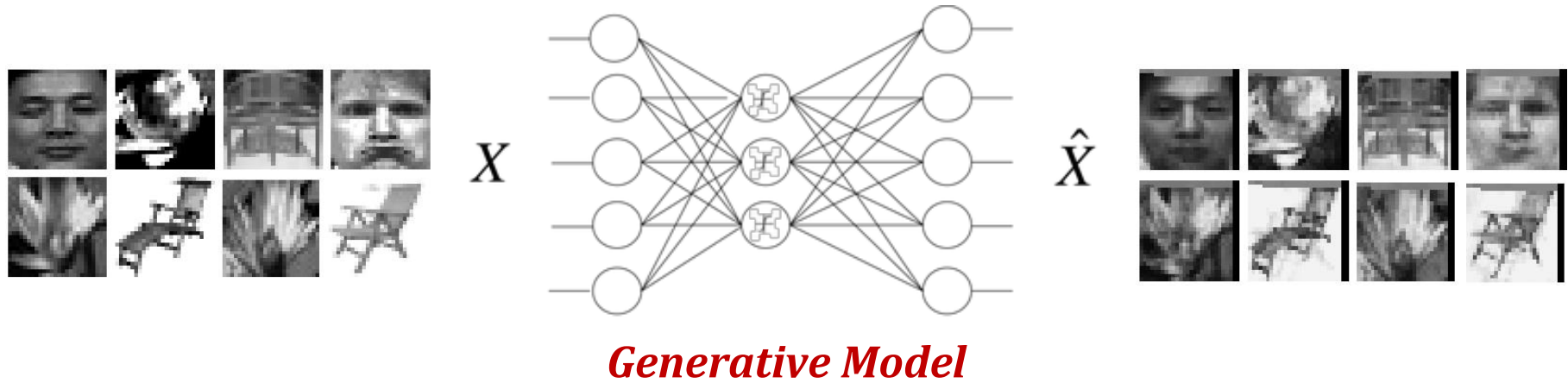
Structured learning



Learning Paradigms



Unsupervised Learning



Pixel RNN

DL for unsupervised learning

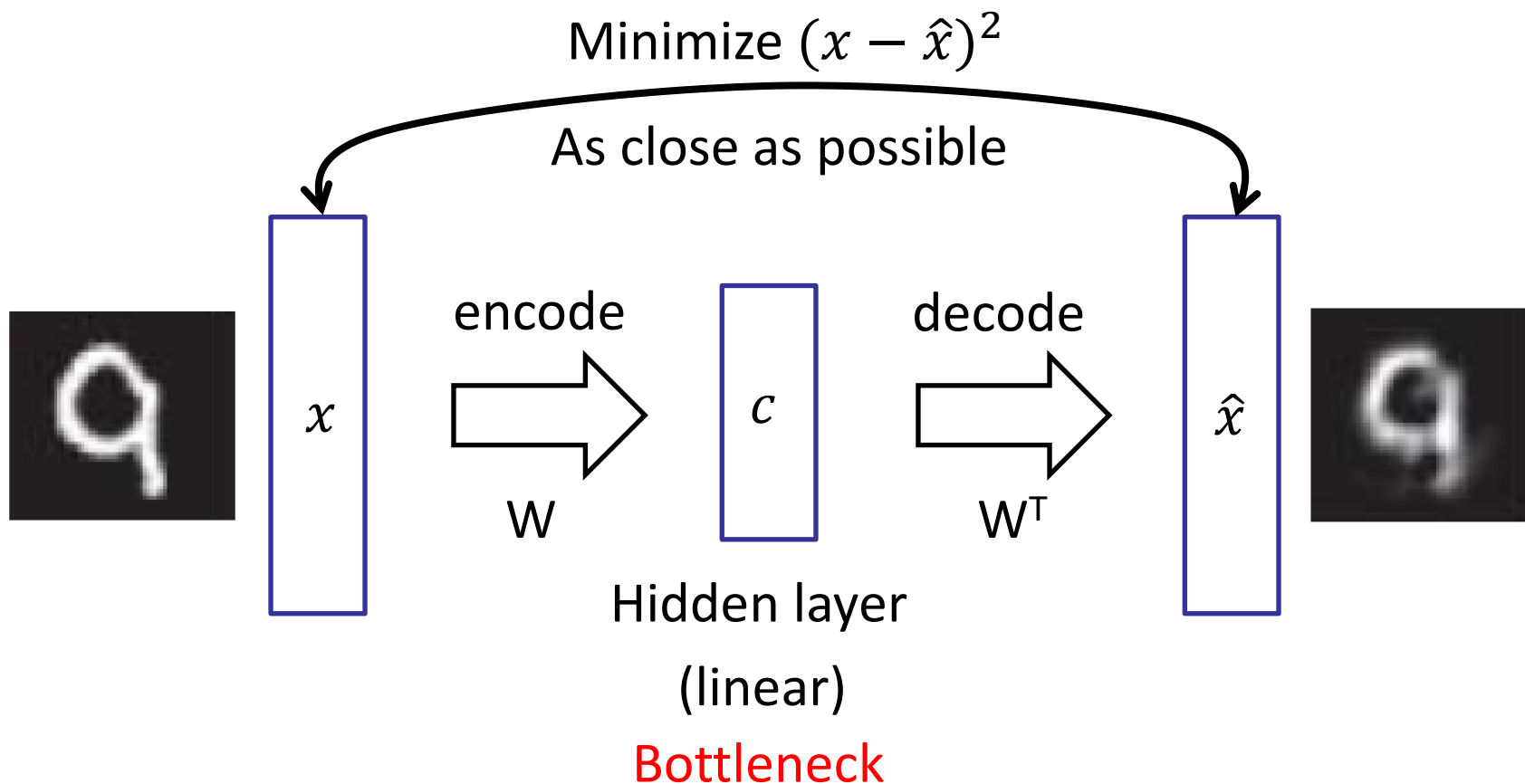
- Generative model: Learning by creating
 - Autoencoder (AE)
 - Variational autoencoder (VAE)
 - Generative adversarial network (GAN)
- Autoregressive: Creating by predicting
 - OpenAI's GPT, a huge language model

what I cannot create, I do not understand.
--- Richard Feynman

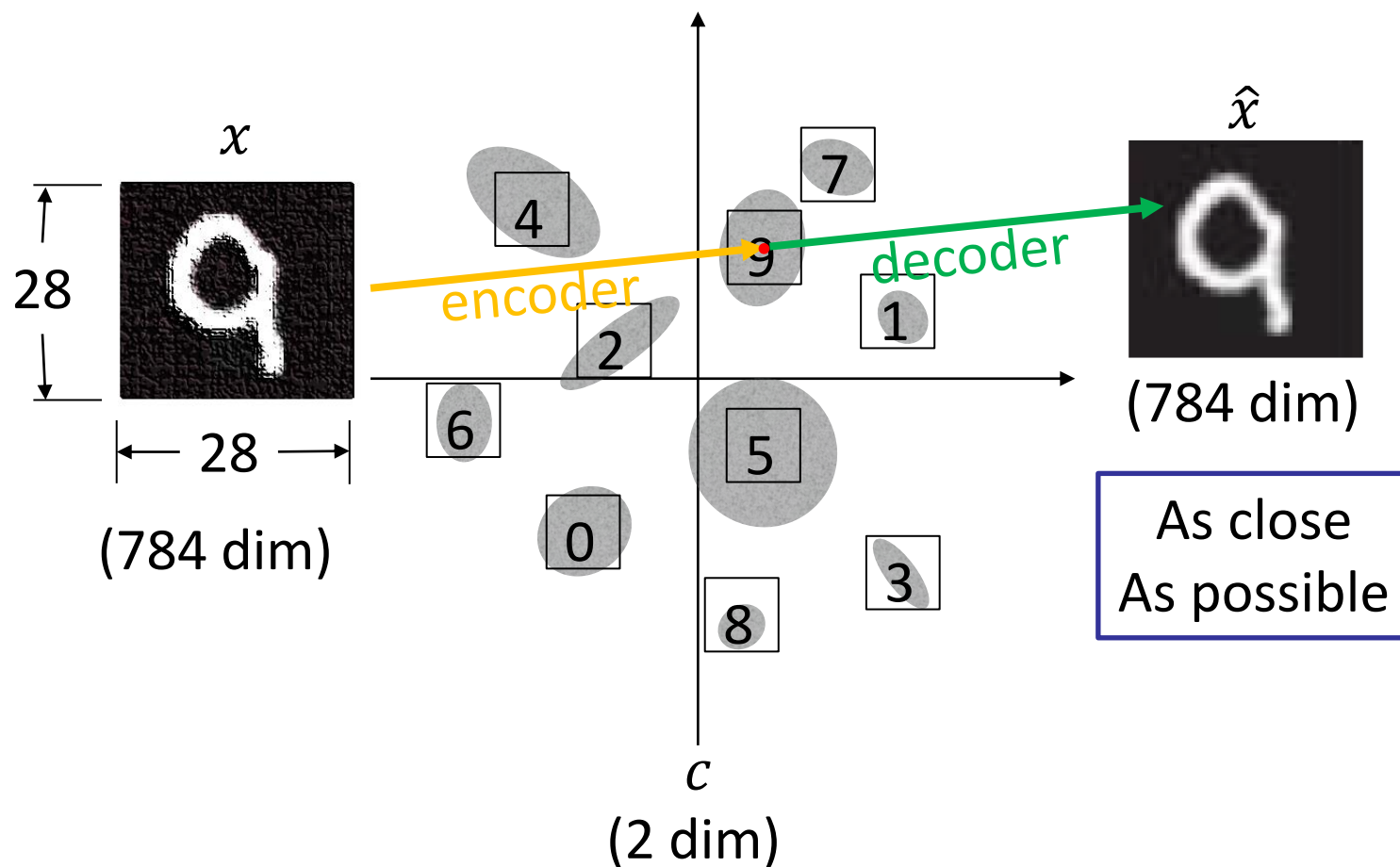
Autoencoder

- Autoencoder

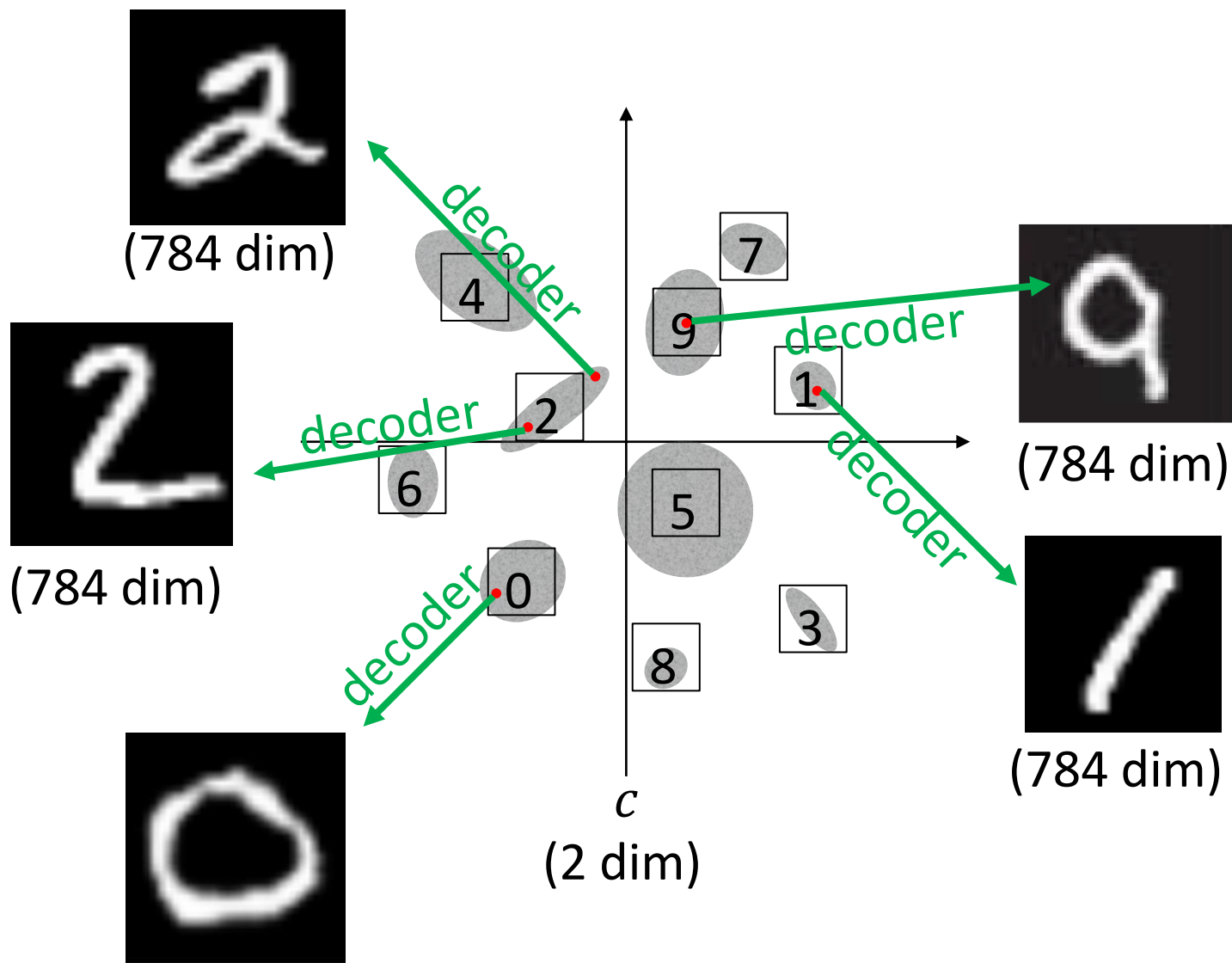
- Use Deep Neural Network to Encode and Decode.



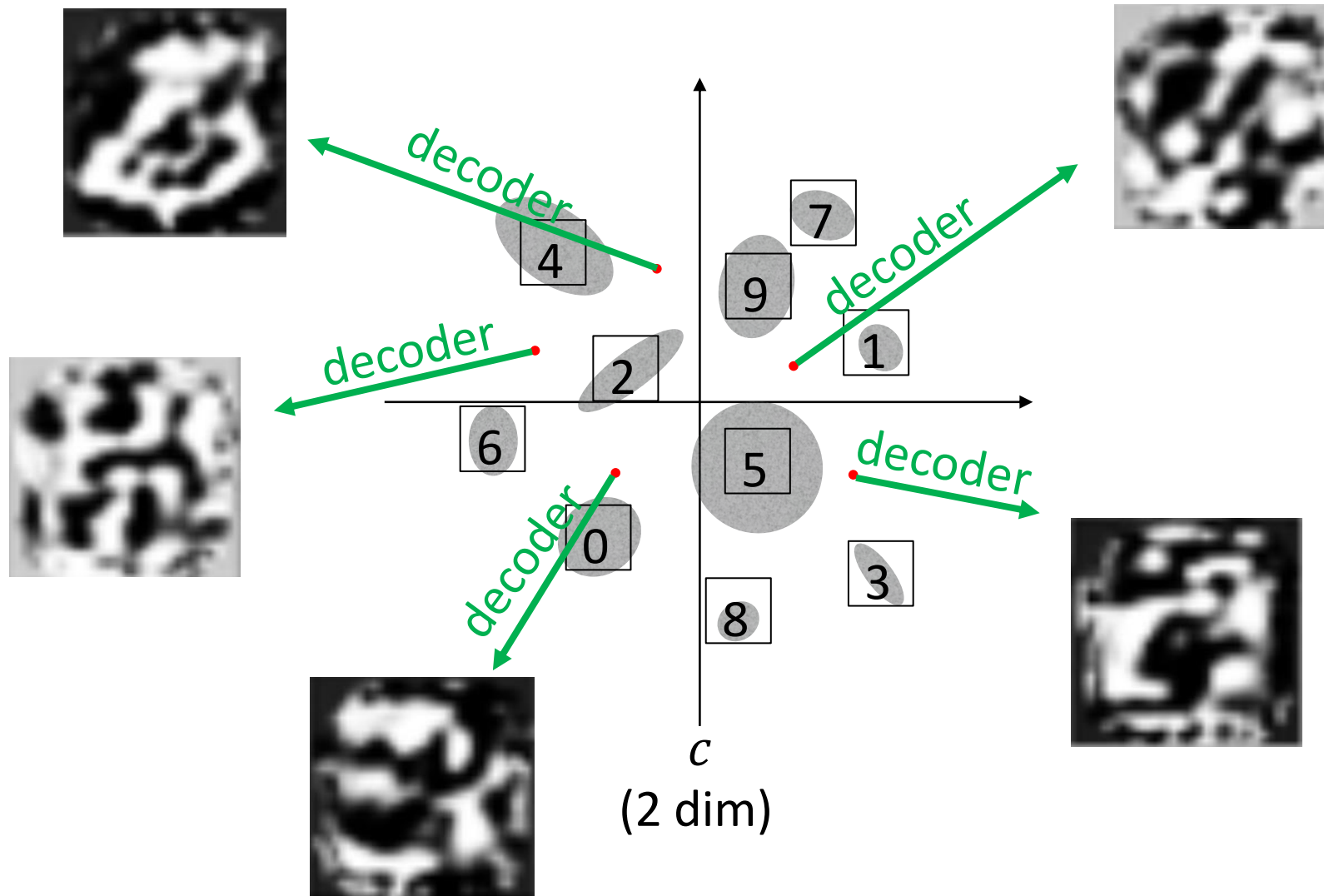
What Autoencoder can do



What Autoencoder can do

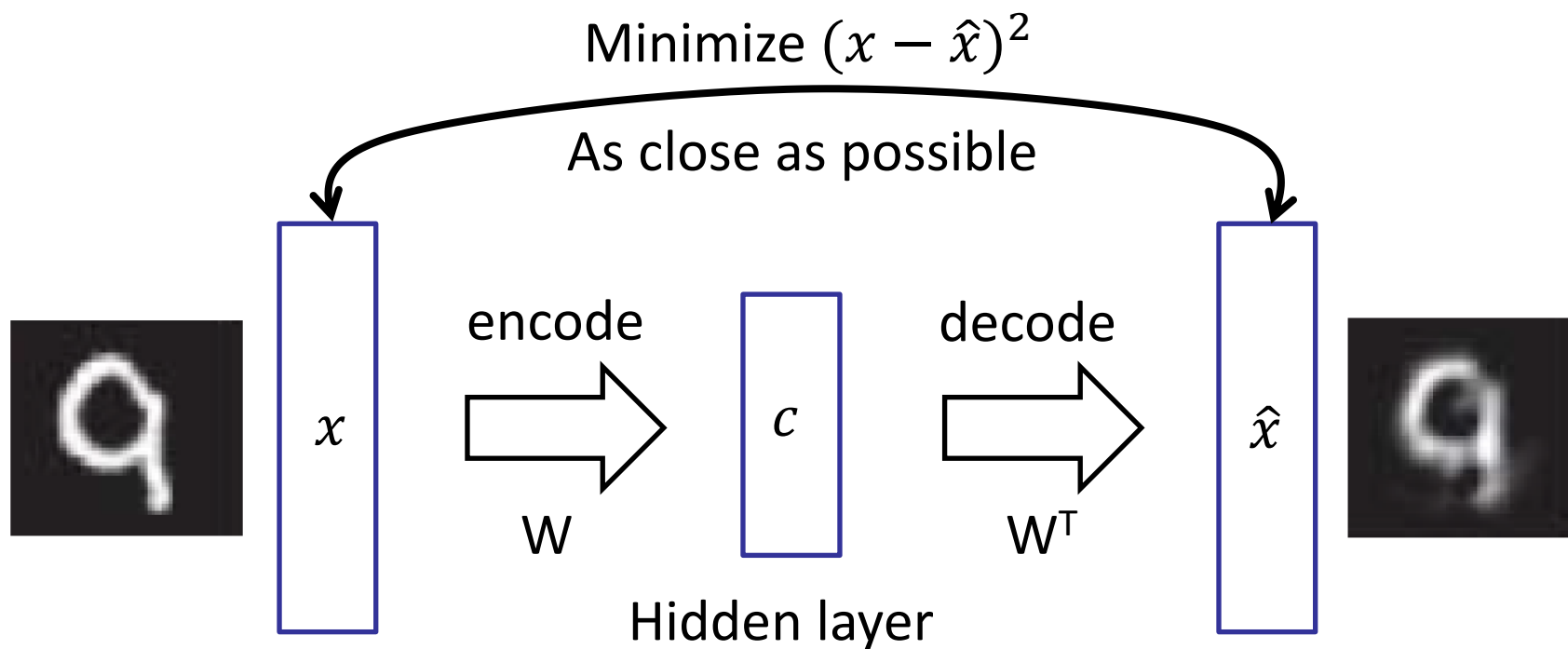


However, Autoencoder cannot do



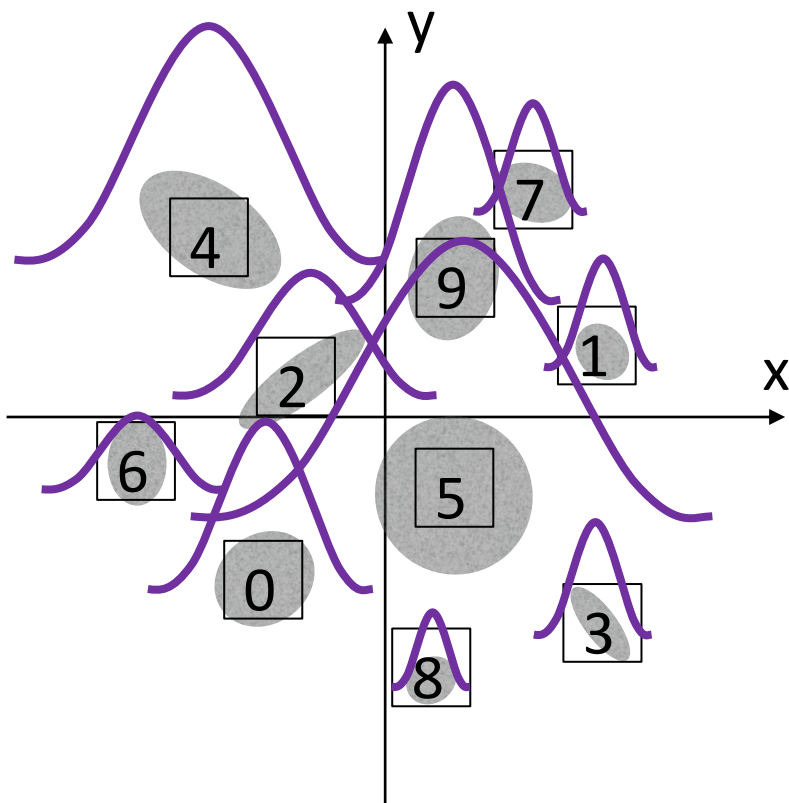
Autoencoder is not good at generating

- We only ask AE to keep the original data closer to the reconstructed data.
- We never tell the algorithm what to do in the middle of two clusters.



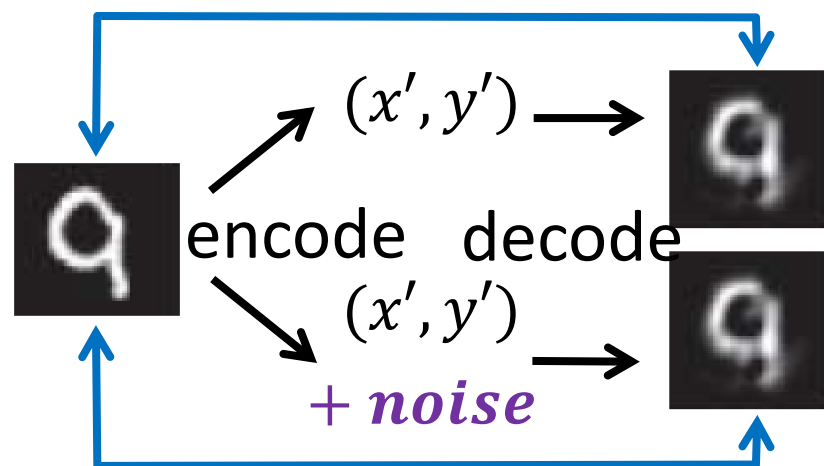
So, what can we do?

—VAE - **V**ariational **A**uto**e**ncoder



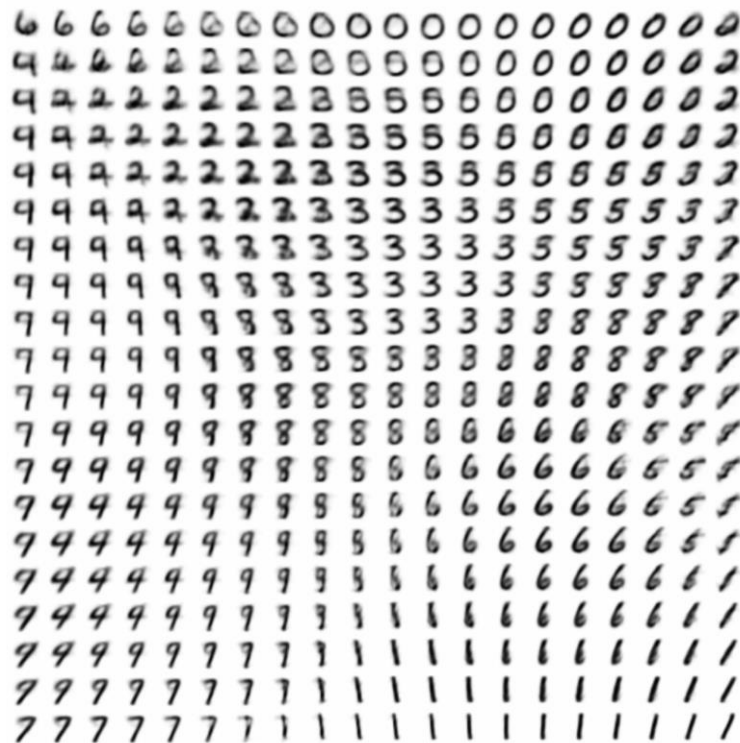
(Gaussian kernel is always a good choice.)

- We add each original data points with some “influence” to other space near it and hope to fill the whole space with probability. Not only keeps these two close



But also can keeps these two close

VAE's results



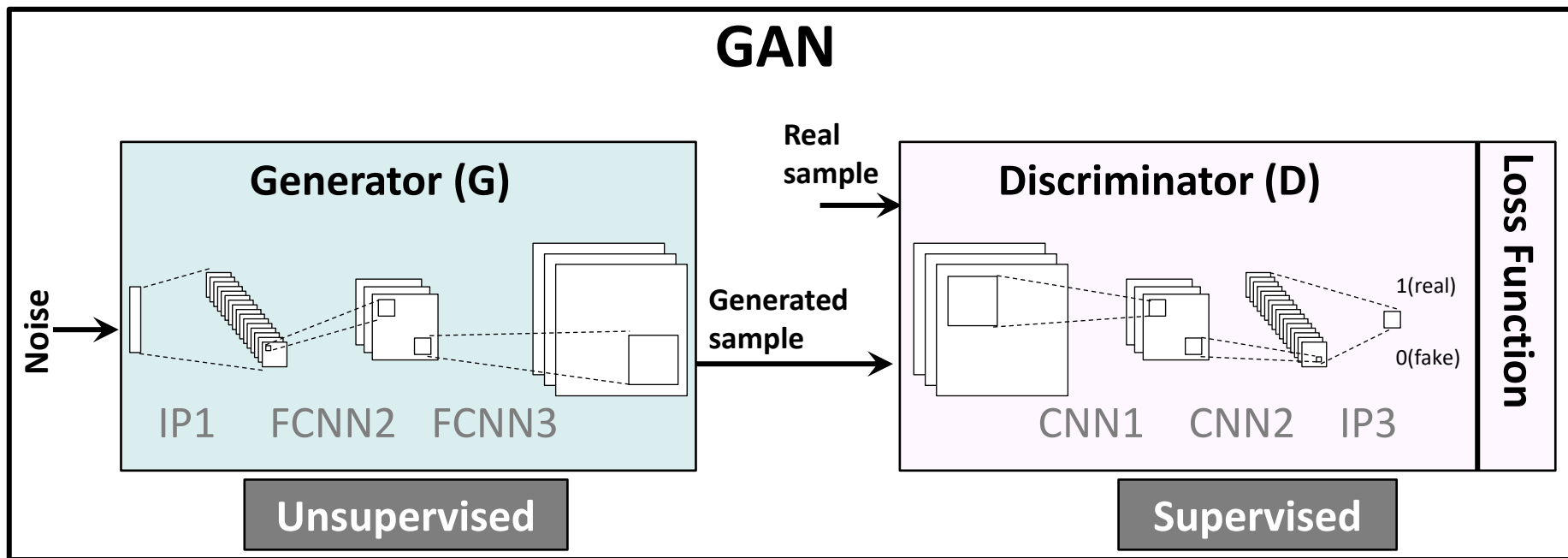
Now:

Data points between two digits looks like digits 😊

Face generating looks good. But not realistic enough.

Generative Adversarial Networks (GAN)

- Two DNNs - a **generator** and a **discriminator** are **co-trained**
- More complex **training procedure** and **data dependency**



GAN – Generative Adversarial Network

- Algorithm
 - Randomly initial generator G and discriminator D .
 - Fix generator G , and update discriminator D .
 - Discriminator learns to assign high scores to real objects and low scores to generated objects
 - Fix D , and update G .
 - Generator learns to “fool” the discriminator.
 - Iterate to step 2 until converged,

Generative Adversarial Networks (GAN)

- Image Generation and Manipulation

- Video Generation and Prediction

- Text Generation

- Photo Inpainting

Generated bedrooms

Video Prediction
Photo Inpainting



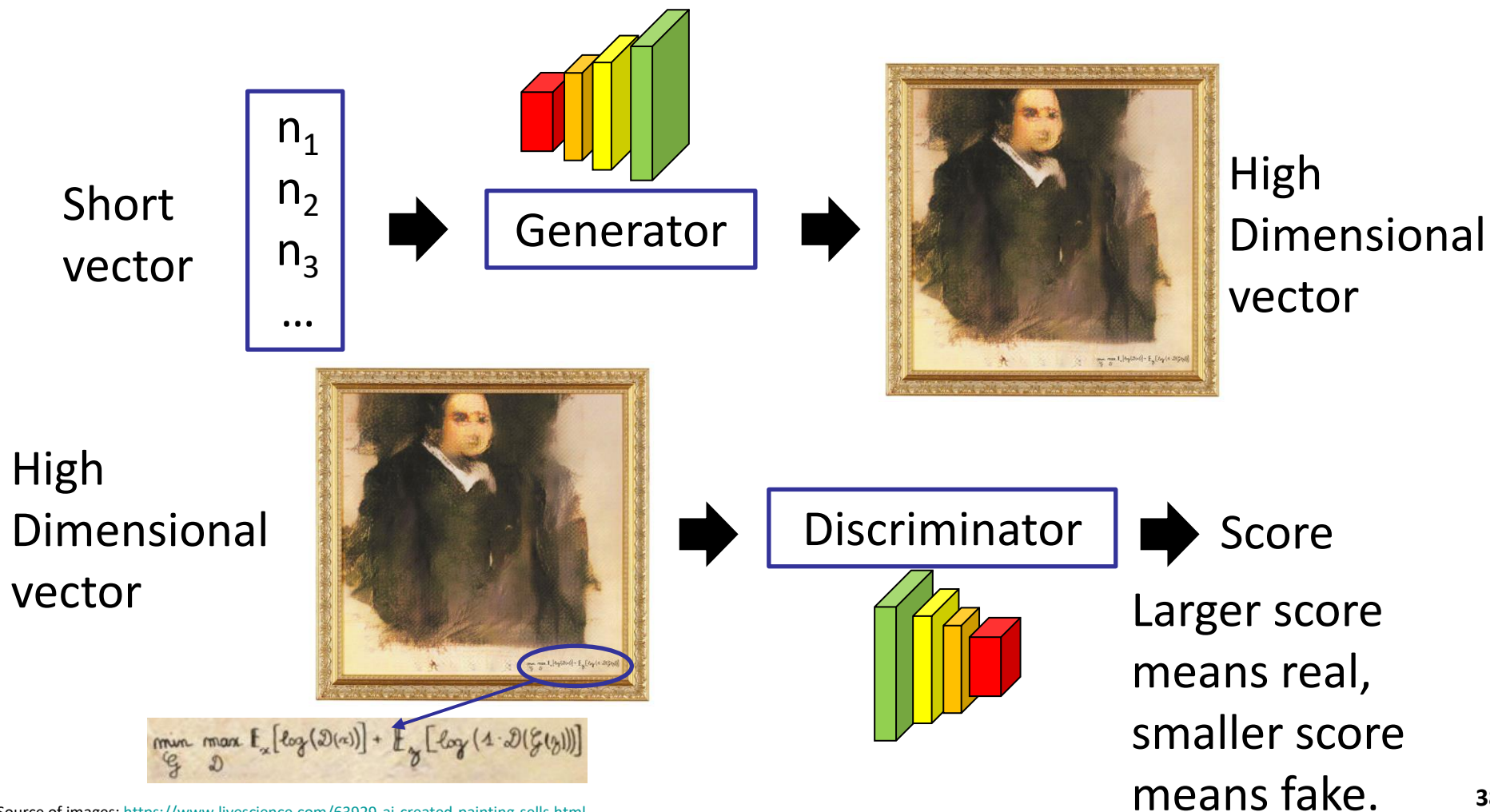
MLE
SeqGAN

A.Radford, et al, 2015.

M. Mathieu et al, 2016

GAN – Generative Adversarial Network

- For images, **the most successful generative model so far** has been the Generative Adversarial Network (GAN)



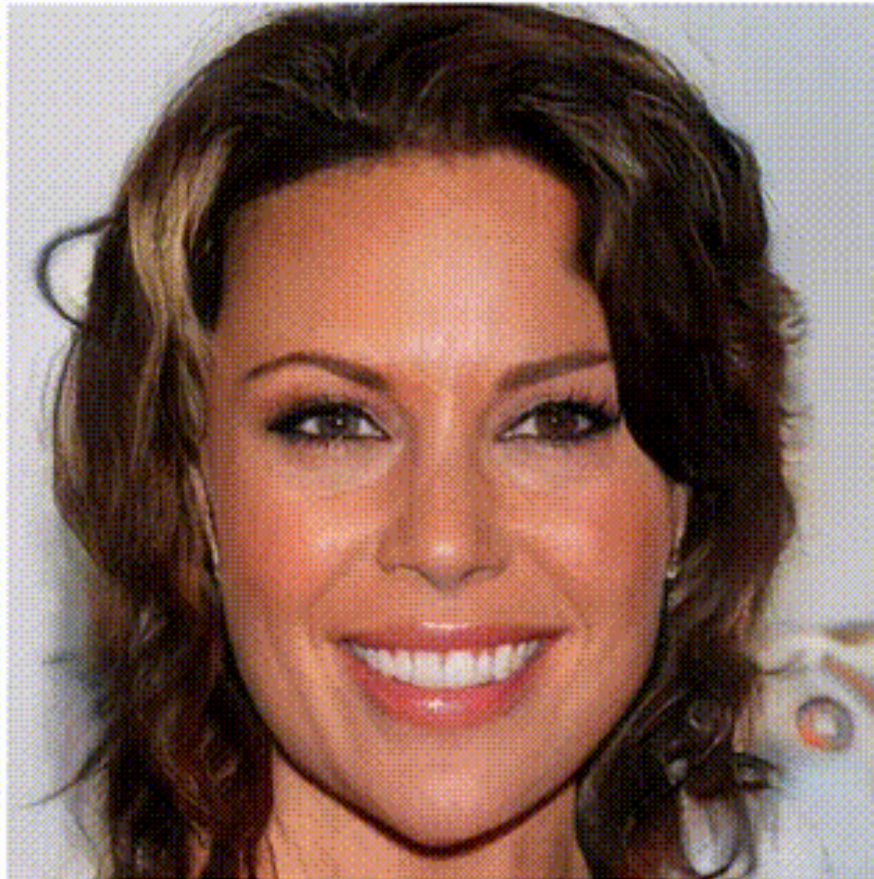
GAN's results

Can you tell which one is a fake image?



GAN's results

INSTRUCTION: press +/- to adjust feature, toggle feature name to lock the feature



random face		
Male	Age	Skin_Tone
- +	- +	- +
Bangs	Hairline	Bald
- +	- +	- +
Big_Nose	Pointy_Nose	Makeup
- +	- +	- +
Smiling	Mouth_Open	Wavy_Hair
- +	- +	- +
Beard	Goatee	Sideburns
- +	- +	- +
Blond_Hair	Black_Hair	Gray_Hair
- +	- +	- +
Eyeglasses	Earrings	Necktie
- +	- +	- +

Text To Image - AI Image Generator

“panda learning math”



<https://deepai.org/machine-learning-model/text2img>

Autoregressive, Creating by predicting.

- The data is split into a sequence of small pieces, each of which is predicted in turn.
- Such models can be used to generate data by successively guessing what will come next, feeding in a guess as input and guessing again.
- OpenAI's GPT
 - Give it a human written start, and it will go on and on. People can use it to write novel story, make fake news, ...

Which could be dangerous

GPT-2's example

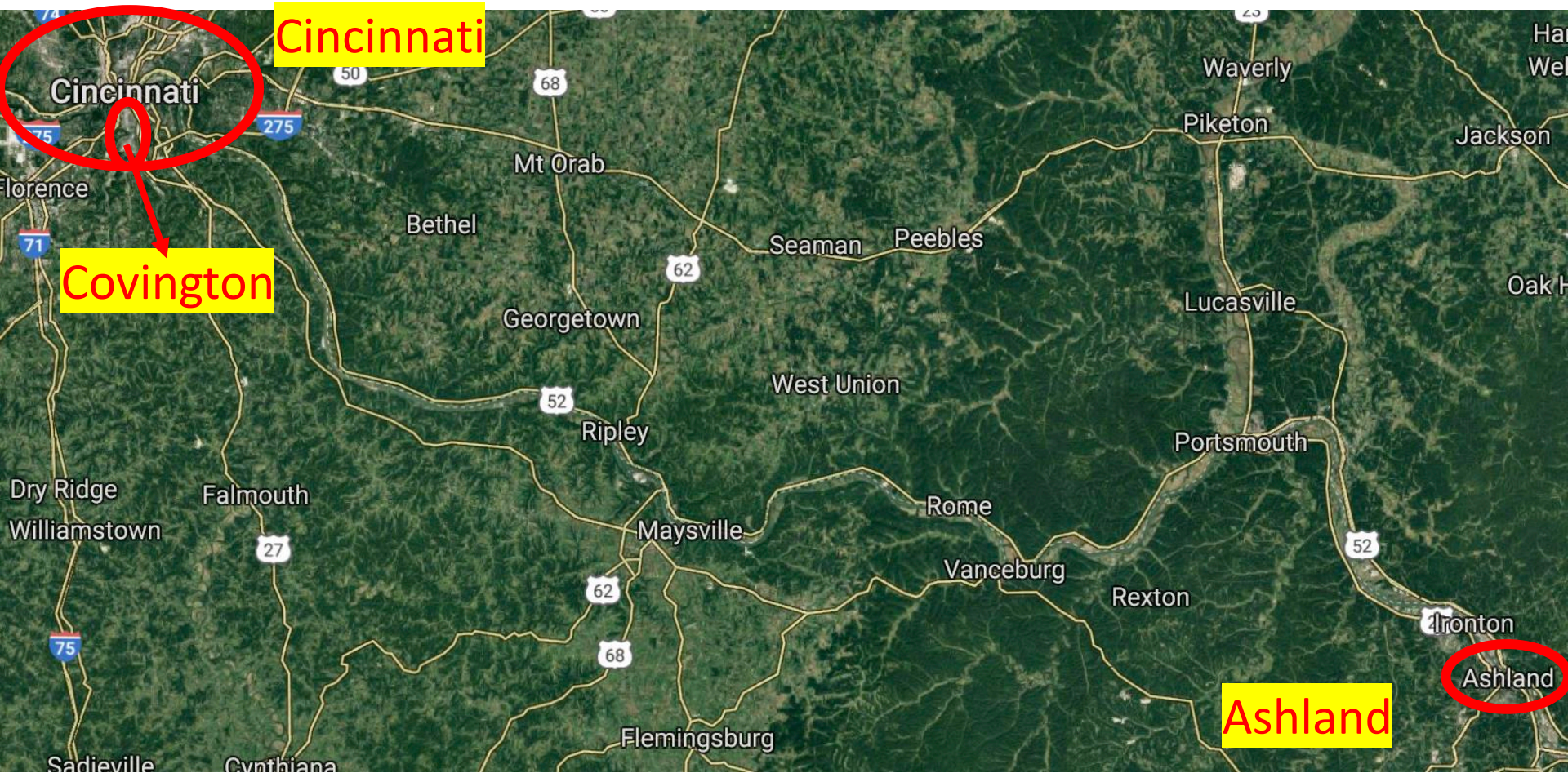
Human written:

A train carriage containing controlled **nuclear materials** was stolen in **Cincinnati** today. Its whereabouts are unknown.

Machine written:

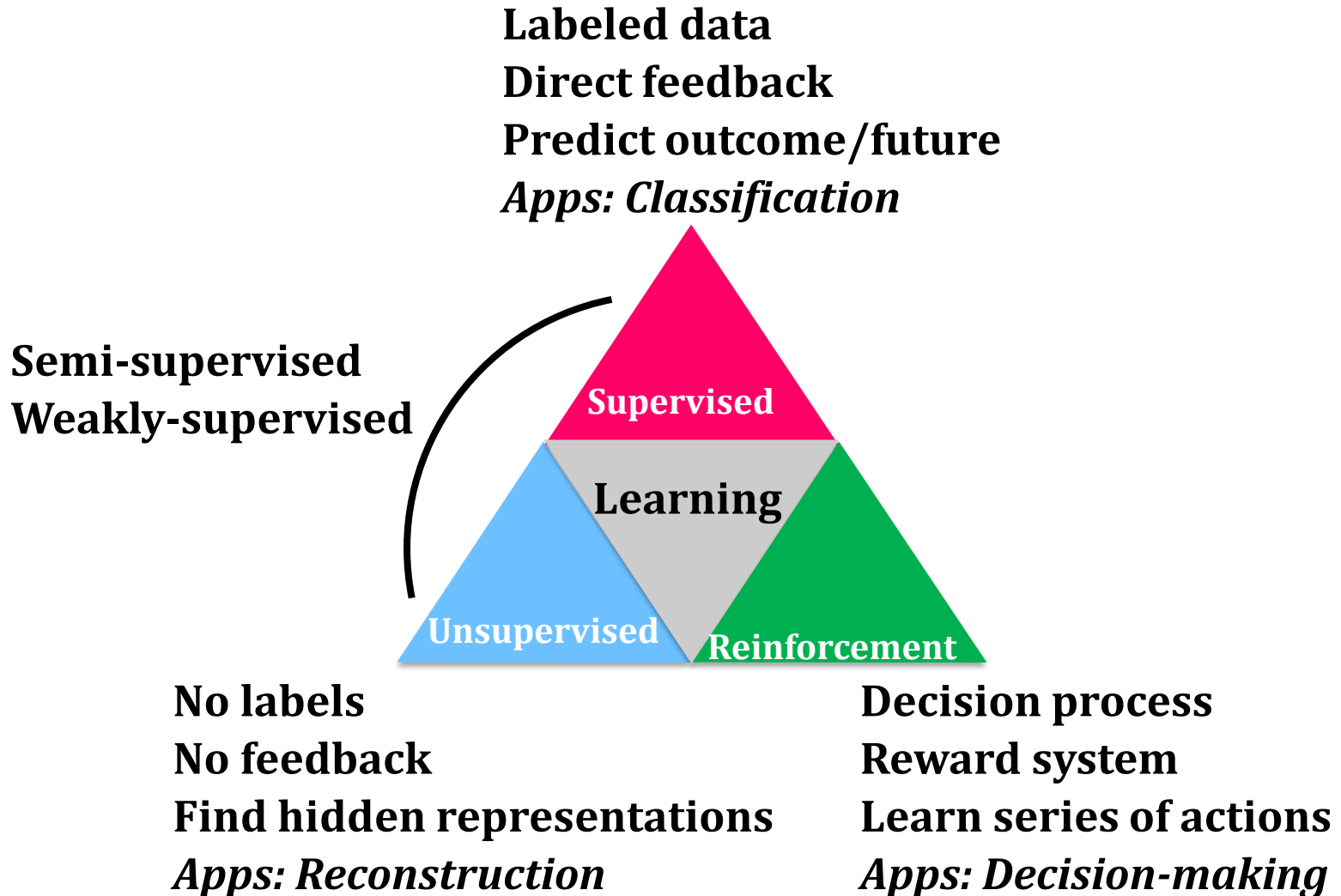
The incident occurred on the downtown train line, which runs from **Covington** and **Ashland** stations. In an email to **Ohio** news outlets, the **U.S. Department of Energy** said it is working with the Federal Railroad Administration to find the thief. “The theft of this nuclear material will have significant negative consequences on public and environmental health, our workforce and the economy of our nation,” said Tom Hicks, the U.S. **Energy Secretary**, in a statement. “Our top priority is to secure the theft and ensure it doesn’t happen again.” The stolen material was taken from the **University of Cincinnati’s** Research Triangle Park **nuclear research site**, according to a news release from Department officials. The **Nuclear Regulatory Commission** ...

GPT-2's example



The news generated automatically seems very **convincible**!

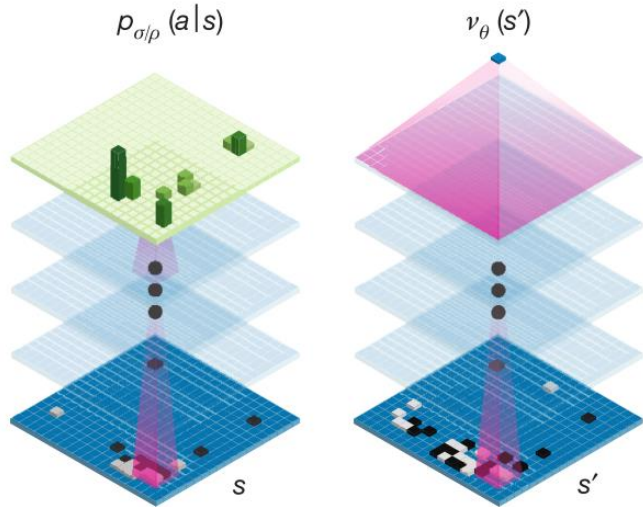
Learning Paradigms



Reinforcement Learning

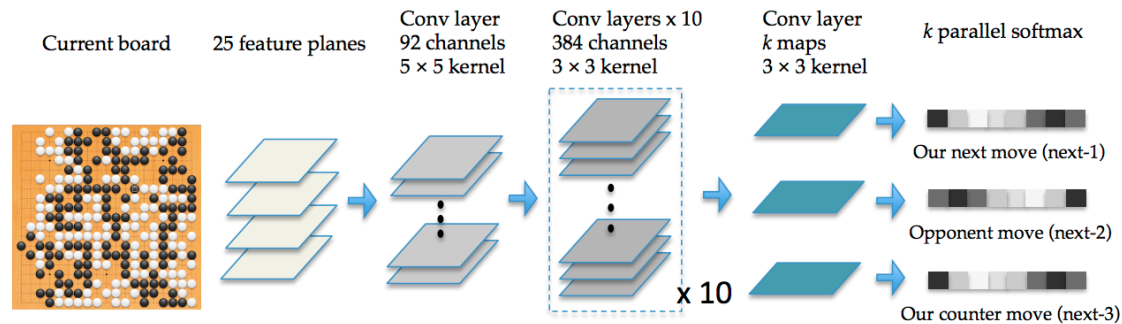
Policy network

Value network

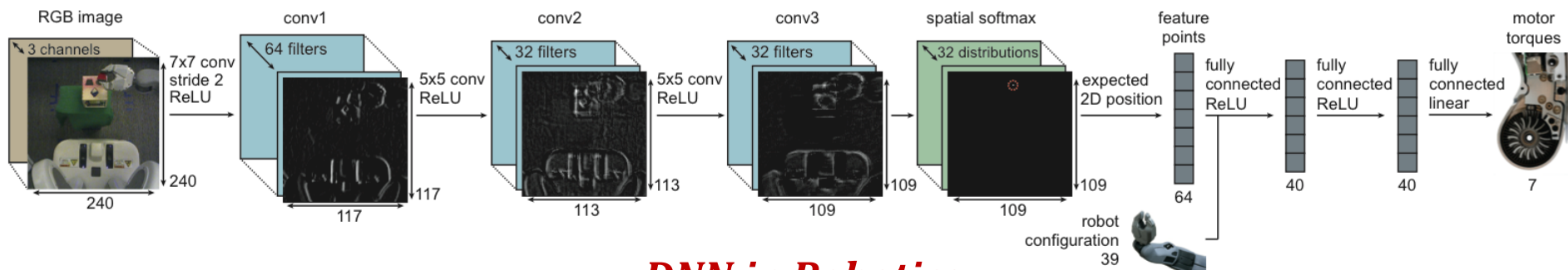


Alpha Go

DarkForest



Facebook Go engine



DNN in Robotics

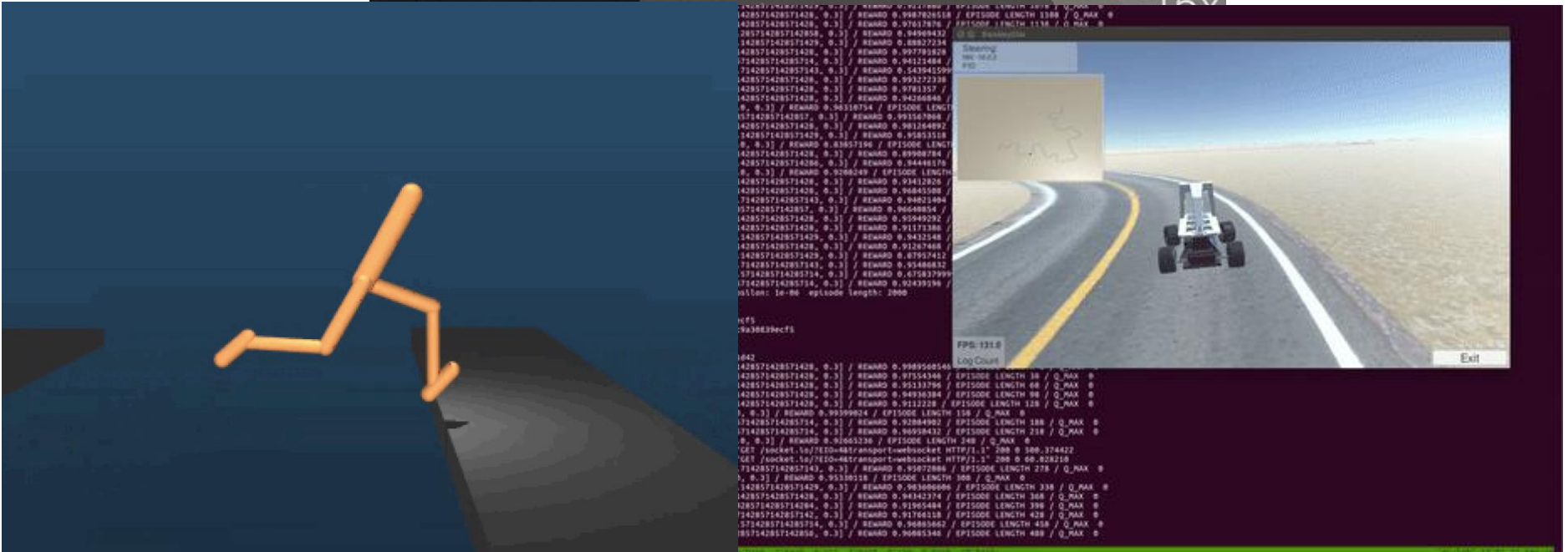
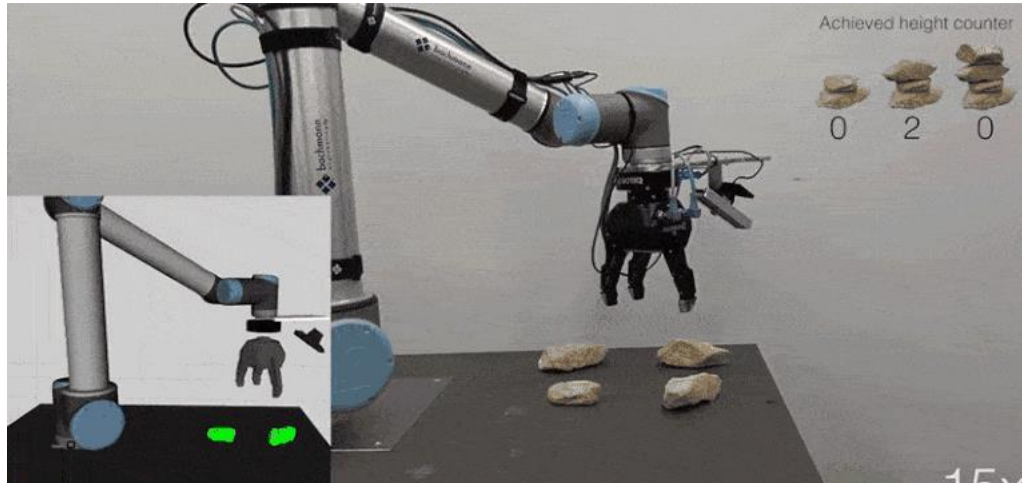
DL for reinforcement learning

- As the number of states increases, the agent's exploration space is increasing exponentially. Traditional methods can no longer support it.
- The function: $f(\text{state}, \text{action}) \rightarrow \text{value}$ will be too difficult and complex to calculate.
- DNNs can fit this function in high-dimensional space.
 - This approach extends reinforcement learning by using a deep neural network and without explicitly designing the state space.

Compare to traditional ML methods

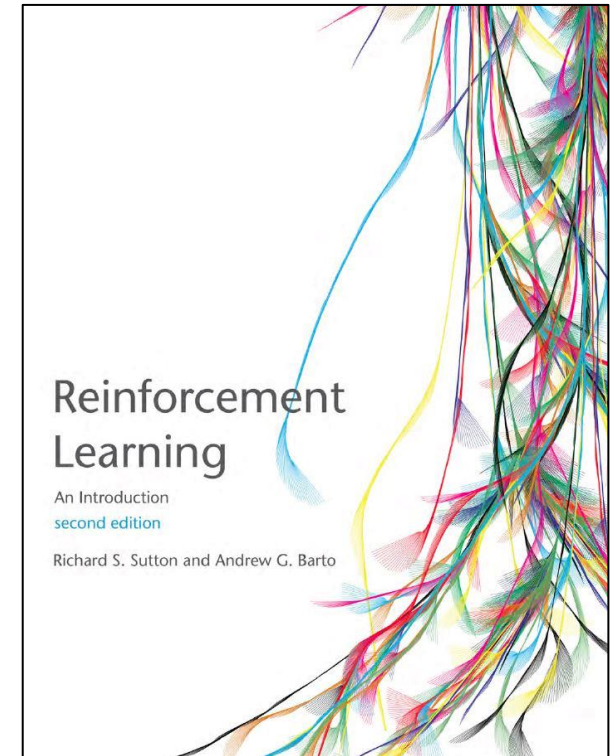
- Traditional Methods
 - Can only play some easy game like back gammon, Inverted pendulum...
- Deep Reinforcement Learning
 - Alpha Go
 - All types of ATARI games

Reinforcement learning examples



Reinforcement learning reference

- Deep mind RL online course
- <https://www.youtube.com/watch?v=TCCjZe0y4Qc&list=PLqYmG7hTraZDVH599EitIEWsUOsJbAodm>



In this chapter, we learned:

- A brief review of the traditional ML methods.
 - Supervised/Unsupervised/Reinforcement learning
- Why we need deep learning
 - Real world: Dimension is high, function is complex.
 - Deep neural network has the capability of doing so.
- DL examples
 - CNN, Structured learning for supervised
 - AE, VAE, GAN, GPT-2 for unsupervised
 - Deep reinforcement learning